

SWIM: Vision-Language-Grounded Soft Whole-Body Interactive Manipulation

Tingcong Liu^{1,4}, Aye Phyu Phyu Aung⁴, Junjie Xiong³, Siyi Ma³, Bo An¹, Ke Wu^{3†}, Senthilnath Jayavelu^{2,4†}

¹Nanyang Technological University, Singapore

²National University of Singapore, Singapore

³Mohamed bin Zayed University of Artificial Intelligence, Abu Dhabi, United Arab Emirates

⁴Institute of Advanced Intelligence and Computing, A*STAR, Singapore

†Corresponding Author

Abstract—Soft and continuum robots enable manipulation through distributed body deformation and contact, yet translating language and visual context into executable whole-body actuation remains a fundamental challenge. We present SWIM, a framework that maps an initial RGB observation and a language instruction to a complete actuation-command sequence. Its vision-language-action (VLA) policy, SWIM-VLA, combines a diffusion action head with Visual Soft Proprioception (VSP) through a shared representation of RGB observations, language instructions, and tendon states. The diffusion head models conditional distributions of expert command chunks, while VSP supervises ordered body-anchor predictions using simulation ground truth, encouraging the representation to retain body geometry when learning from limited demonstrations. Embodied mechanical intelligence supports physical execution of command sequences generated through iterative virtual rollout from evolving simulated observations, with intrinsic compliance providing local contact adaptation without online policy queries. We evaluate SWIM on packing, reaching, and grasping on a planar tendon-driven soft robot, with grasping targets anchored. In simulation, SWIM-VLA achieves success rates of 100%, 96%, and 88%, respectively, outperforming an adapted OpenVLA-OFT baseline and controlled ablations. On hardware, SWIM achieves success rates of 100%, 80%, and 75%, compared with 75%, 40%, and 25% for direct online deployment of the same policy checkpoint.

I. INTRODUCTION

Soft and continuum robots combine structural compliance with large deformations, supporting applications in object handling, operation in confined spaces, and minimally invasive procedures [1], [2]. In object manipulation, their deformable bodies enable whole-body wrapping and enclosure through distributed contact [3], [4]. Research on this capability has pursued two complementary directions: regulating body shape and configuration [5], [6], and coordinating body motion and contact for manipulation [7], [8].

For configuration control, Pustina et al. [5] developed feedback laws to regulate prescribed soft-robot shapes under model uncertainty and constant disturbances, while Almanzor et al. [6] used deep visual inverse kinematics to reach user-specified full-body target shapes. Beyond configuration control, early OctArm experiments demonstrated teleoperated whole-arm grasping and object handling [3]. Marchese et al. [7] planned whole-arm motion while accounting for the changing body envelope in confined environments. Li and Xiao [4] planned probing and progressive contact formation for whole-arm grasping in cluttered environments. More

recent work combined distributed tactile sensing with hierarchical control for autonomous underwater grasping [9], while Yang et al. [10] learned whole-body grasping from actuation-space demonstrations. These studies primarily address shape regulation and manipulation once a desired configuration or task-specific objective has been specified. Instruction-driven whole-body manipulation introduces an additional task-grounding requirement: the robot must interpret the requested behavior and identify task-relevant targets from language and observed scenes. This requires connecting semantic intent and visual context to physically executable whole-body actuation.

Vision-language-action (VLA) models map language instructions and visual observations to robot actions. Established formulations have largely been developed for rigid manipulators, using Cartesian end-effector motions, gripper states, or joint commands [11], [12], [13]. Recent work has extended vision-language-conditioned action generation to soft and continuum robots. These studies address end-effector-centered object manipulation [14], [15], target tracking and navigation using discrete motor commands [16], [17], and the generation of motion primitives and expressive whole-body behaviors through low-level actuation [18], [19]. Extending these advances to manipulation through distributed body contact requires addressing three mismatches. First, an *action-interface mismatch* arises because end-effector motion does not fully specify the coordinated body deformation and contact required for manipulation [8]. A given task condition may also admit multiple valid actuation sequences, motivating policies that capture conditional action distributions [10], [20]. Second, a *whole-body state mismatch* arises because end-effector-centered state descriptions do not fully capture the high-dimensional body configurations that govern soft whole-body interactions [15]. Moreover, action targets provide only indirect supervision of body geometry, making the relationship between body deformation and actuation difficult to learn from limited demonstrations. Third, an *execution mismatch* arises from both sim-to-real discrepancies and delays in online deployment. For policies trained in simulation, visual differences can alter policy inputs, while inaccuracies in robot and contact models can change the physical response to predicted commands [21], [10]. Inference and communication latency can further cause commands to be applied after the body configuration and

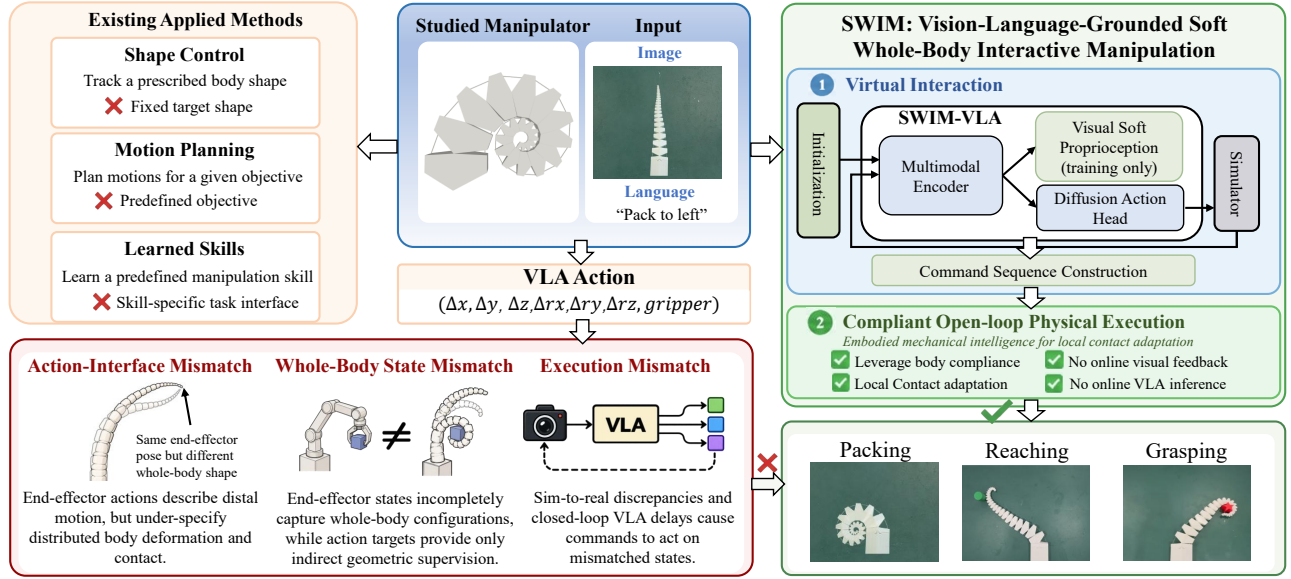


Fig. 1. Motivation and design principles of SWIM for soft whole-body manipulation.

contact state have changed [13], [22].

Together, these mismatches complicate the translation of language and visual context into whole-body manipulation, particularly when demonstrations are limited and physical execution differs from the conditions assumed during learning. Motivated by these challenges, we introduce SWIM, a framework for vision-language-grounded soft whole-body interactive manipulation. Our contributions are as follows:

- **SWIM: virtual-rollout deployment supported by embodied mechanical intelligence.** SWIM generates a complete actuation-command sequence from an initial RGB observation and language instruction through virtual rollouts using our proposed SWIM-VLA policy. Intrinsic body compliance supports open-loop execution through local contact adaptation, reducing centralized feedback demands while keeping VLA inference outside the physical execution loop.
- **SWIM-VLA: learning actuation distributions with geometric supervision.** We introduce SWIM-VLA, a tendon-state-conditioned policy combining a diffusion action head with Visual Soft Proprioception (VSP). The diffusion action head models the conditional distribution of expert command chunks, capturing alternative whole-body actuation strategies under the same task conditions. VSP uses ground-truth body-anchor coordinates from simulation to encourage the shared representation to retain body geometry, supporting learning from limited demonstrations.
- **Evaluation of policy design and physical deployment.** We evaluate packing, reaching, and grasping tasks on a planar tendon-driven robot, with targets anchored during grasping. Simulation ablations assess VSP and diffusion-based action prediction, while physical experiments compare virtual-rollout execution with direct

online deployment of the same policy checkpoint.

II. PROBLEM STATEMENT

A. Objective

We study vision-language-grounded whole-body manipulation for soft robots. A task is specified by an initial RGB observation $\mathbf{I}$ of the robot and task scene, together with a language instruction ℓ . The desired output is a complete actuation-command sequence

$$\mathbf{U} = [\mathbf{a}_0, \dots, \mathbf{a}_{T-1}]^\top \in \mathbb{R}^{T \times N_a}, \quad (1)$$

where T is the number of command steps in the complete sequence, N_a is the number of independently controlled actuation channels, and $\mathbf{a}_t \in \mathbb{R}^{N_a}$ is the command vector at step t . We seek a sequence whose physical execution completes the language-specified task through whole-body deformation and, where required, distributed contact.

B. The studied manipulator

We evaluate the formulated problem on SpiRob, a tendon-driven spiral continuum robot [23], as shown in Fig. 1. We use a planar SpiRob with a 24-joint logarithmic-spiral body actuated by two independently driven tendons. Its monolithic compliant structure enables large and continuous whole-body deformation despite the low-dimensional actuation space. Through the same actuation interface, SpiRob can produce diverse behaviors, including extension, retraction, bending, curling, wrapping, and enclosure, making it well suited for studying coordinated whole-body deformation and distributed contact.

III. METHOD

SWIM maps an initial RGB observation $\mathbf{I}$ and language instruction ℓ to a complete actuation-command sequence $\mathbf{U}$. As illustrated in Fig. 2, the framework consists of two stages.

SWIM: Vision-Language-Grounded Soft Whole-Body Interactive Manipulation

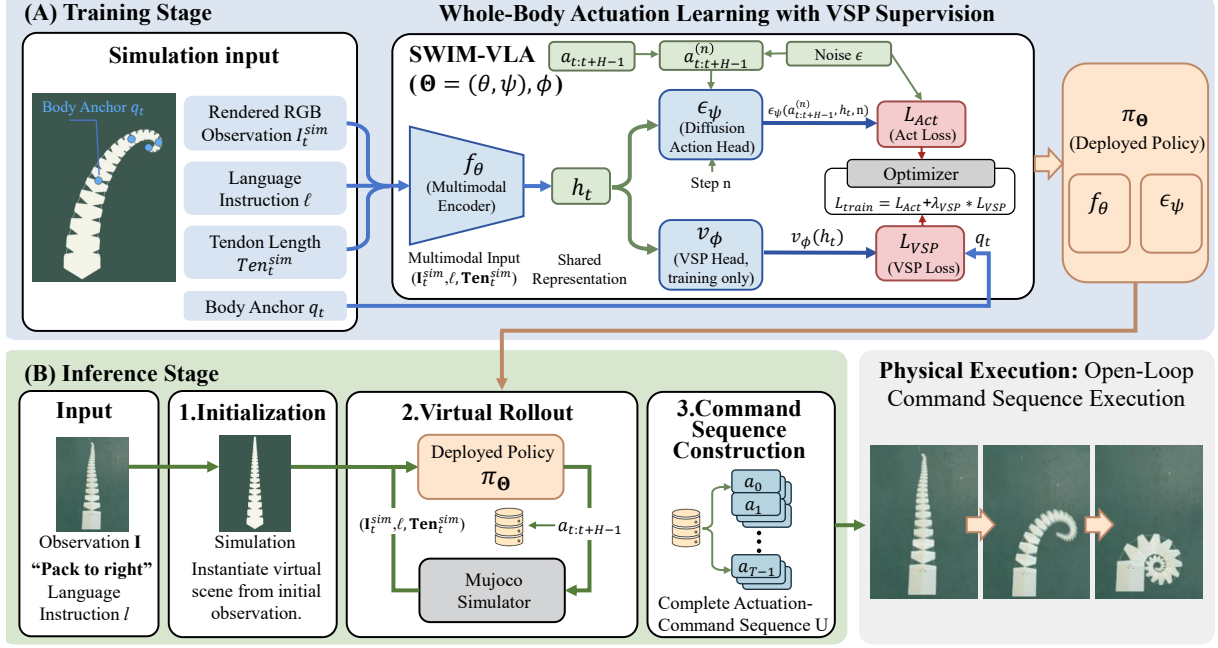


Fig. 2. Overview of the SWIM framework. (A) SWIM-VLA is trained from simulated interaction trajectories using diffusion action learning with VSP supervision. (B) At inference, the trained policy performs virtual rollout to construct a complete actuation-command sequence, which is then executed open-loop on the physical robot.

During training, SWIM-VLA learns to predict command chunks through a diffusion action head from simulated whole-body interaction trajectories with VSP supervision. During inference, SWIM uses the trained policy to construct U through virtual rollout, and the resulting sequence is executed open-loop on the physical robot.

A. SWIM-VLA Training from Simulated Demonstrations

SWIM-VLA builds on OpenVLA-OFT [13], using the current tendon-length vector as a proprioceptive input and representing policy actions as actuation commands. As shown in Fig. 2(A), SWIM-VLA is trained on simulated whole-body interaction demonstrations. At simulated command step t , the policy predicts an H -step actuation-command chunk from the multimodal input:

$$\mathbf{a}_{t:t+H-1} \sim \pi_\Theta(\cdot | \mathbf{I}_t^{\text{sim}}, \ell, \text{Ten}_t^{\text{sim}}). \quad (2)$$

Here, $\mathbf{I}_t^{\text{sim}}$ is the rendered RGB observation, ℓ is the language instruction, and $\text{Ten}_t^{\text{sim}} \in \mathbb{R}^{N_a}$ is the current tendon-length vector. The tendon-length vector provides a low-dimensional proprioceptive cue to the robot's current whole-body configuration. The output $\mathbf{a}_{t:t+H-1} \in \mathbb{R}^{H \times N_a}$ contains H actuation commands for the N_a tendon channels. We denote the deployable policy parameters by $\Theta = (\theta, \psi)$.

1) *Shared Multimodal Representation*: The multimodal input is first encoded as

$$\mathbf{h}_t = f_\theta(\mathbf{I}_t^{\text{sim}}, \ell, \text{Ten}_t^{\text{sim}}), \quad (3)$$

where f_θ is the multimodal encoder and $\mathbf{h}_t$ is the shared multimodal representation. As depicted in Fig. 2(A), $\mathbf{h}_t$ is

passed to the diffusion action head ϵ_ψ for command-chunk prediction and to the auxiliary VSP head v_ϕ for whole-body geometric supervision.

2) *Actuation-Space Diffusion Objective*: Prior work on the soft robot has shown that different sampled actuation-command sequences can achieve the same whole-body grasping task [10], suggesting that a task condition may admit multiple feasible actuation realizations. Motivated by this distributional structure, we adopt a diffusion action head [20], [13] to model the conditional distribution over H -step command chunks spanning all actuation channels.

Given the shared multimodal representation $\mathbf{h}_t$, a demonstrated command chunk is perturbed with Gaussian noise, and ϵ_ψ is trained to predict the added noise:

$$\begin{aligned} \mathbf{a}_{t:t+H-1}^{(n)} &= \sqrt{\bar{\alpha}_n} \mathbf{a}_{t:t+H-1} + \sqrt{1 - \bar{\alpha}_n} \epsilon, \\ \mathcal{L}_{\text{act}}(\theta, \psi) &= \mathbb{E}_{t,n,\epsilon} \left[\left\| \epsilon - \epsilon_\psi \left(\mathbf{a}_{t:t+H-1}^{(n)}, n, \mathbf{h}_t \right) \right\|_2^2 \right]. \end{aligned} \quad (4)$$

Here, θ and ψ parameterize the multimodal encoder f_θ and diffusion action head ϵ_ψ , respectively. $\mathbf{a}_{t:t+H-1}^{(n)}$ denotes the noisy command chunk at diffusion step n , $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ is the sampled Gaussian noise, and $\bar{\alpha}_n$ is the cumulative noise-schedule coefficient at step n .

3) *Visual Soft Proprioception*: End-effector states do not capture the full whole-body configuration, while tendon lengths and action targets provide only indirect geometric cues. We therefore introduce Visual Soft Proprioception (VSP), a training-time auxiliary objective that supervises spatially distributed body anchors and encourages $\mathbf{h}_t$ to re-

tain whole-body geometric information relevant to command prediction.

As illustrated in Fig. 2 (A), let $\mathbf{q}_t \in [0, 1]^{N_b \times 2}$ denote the normalized image-plane coordinates of N_b body anchors ordered along the backbone from proximal to distal. Under the limited-demonstration setting, these distributed anchors provide a compact geometric descriptor covering both intermediate body locations and the distal region. The VSP head v_ϕ predicts the anchor coordinates using the following objective function:

$$\mathcal{L}_{\text{VSP}}(\theta, \phi) = \mathbb{E}_t \left[\frac{1}{N_b} \sum_{j=1}^{N_b} \|[v_\phi(\mathbf{h}_t)]_j - \mathbf{q}_{t,j}\|_2 \right]. \quad (5)$$

Here, $[v_\phi(\mathbf{h}_t)]_j$ and $\mathbf{q}_{t,j}$ denote the predicted and ground-truth image-plane coordinates of the j -th body anchor, respectively, and the loss averages the Euclidean localization error over the N_b ordered anchors. By supervising multiple locations along the deformable body, VSP encourages $\mathbf{h}_t$ to retain spatially distributed geometric information useful for command-chunk prediction.

4) *Joint Training Objective:* The two objectives are jointly optimized as

$$\mathcal{L}_{\text{train}}(\theta, \psi, \phi) = \mathcal{L}_{\text{act}}(\theta, \psi) + \lambda_{\text{VSP}} \mathcal{L}_{\text{VSP}}(\theta, \phi), \quad (6)$$

where $\lambda_{\text{VSP}} \geq 0$ controls the contribution of the VSP supervision. During training, the multimodal encoder parameters θ are optimized by both objectives, while ψ and ϕ are optimized by $\mathcal{L}_{\text{act}}$ and $\mathcal{L}_{\text{VSP}}$, respectively. After training, the auxiliary VSP head v_ϕ is discarded, and f_θ and ϵ_ψ define the trained SWIM-VLA policy π_Θ .

B. Command-Sequence Construction via Virtual Rollout

As illustrated in Fig. 2(B), SWIM constructs a complete command sequence through iterative virtual rollouts with the trained SWIM-VLA policy. The collected commands are then executed open-loop on the physical robot, where intrinsic body compliance supports local contact adaptation.

1) *Virtual Scene Initialization:* Given the initial RGB observation $\mathbf{I}$, SWIM initializes a virtual scene using pre-defined object models and calibrated planar locations [24]. All objects are instantiated and included in the rendered observation, allowing SWIM-VLA to predict target-directed commands from the scene and language instruction ℓ during virtual rollout. The simulated robot starts from the same pre-defined configuration used for physical reset. The resulting initial policy input is

$$\text{Init}(\mathbf{I}, \ell) = (\mathbf{I}_0^{\text{sim}}, \ell, \mathbf{Ten}_0^{\text{sim}}), \quad (7)$$

where $\mathbf{I}_0^{\text{sim}}$ is the rendered RGB observation of the initialized scene, and $\mathbf{Ten}_0^{\text{sim}}$ is the tendon-length vector determined by the reset configuration.

2) *Virtual Rollout:* Starting from the initialized virtual scene, SWIM repeatedly queries the trained SWIM-VLA policy defined in Eq. (2). At each query, the policy predicts an H -step actuation-command chunk, and the first K

commands are executed in simulation:

$$\mathbf{a}_t^{\text{exec}} = \mathbf{a}_{t:t+K-1}, \quad K \leq H. \quad (8)$$

The resulting RGB observation and tendon-length vector are then used for the next policy query.

3) *Command Sequence Construction:* As shown in Fig. 2(B), during virtual rollout, each executed command segment is appended in temporal order to the command sequence:

$$\mathbf{U} \leftarrow \text{concat}(\mathbf{U}, \mathbf{a}_t^{\text{exec}}). \quad (9)$$

Repeating this process until task completion yields the complete actuation-command sequence

$$\mathbf{U} = [\mathbf{a}_0, \dots, \mathbf{a}_{T-1}]^\top, \quad (10)$$

where T is the number of command steps in the complete sequence. The sequence is then executed open-loop on the physical robot without online queries to SWIM-VLA. This deployment scheme relies on embodied mechanical intelligence: intrinsic body compliance accommodates local contact variations during physical execution, while iterative VLA inference remains within the virtual scene.

IV. EXPERIMENTAL VALIDATION

We evaluate the proposed framework from two aspects: (1) **In simulation**, we evaluate SWIM-VLA on packing, reaching, and grasping tasks, together with controlled ablations of VSP and the diffusion action head; (2) **In physical deployment**, we evaluate SWIM by comparing its virtual-rollout-based command-sequence execution with the direct online execution of the same trained SWIM-VLA policy.

A. Experimental Setup and Evaluation Protocol

1) *Platform and Implementation:* Experiments are conducted using the SpiRob platform described in Sec. II-B. We use matched robot geometry, tendon-actuation interfaces, and top-down monocular observation configurations in MuJoCo simulation and on the physical platform. SWIM-VLA uses OpenVLA-OFT as its backbone. We jointly fine-tune a single policy on simulated demonstrations across different tasks. The principal implementation and training settings are summarized in Table I.

2) *Tasks and Success Criteria:* Following the characteristic motion progression demonstrated by SpiRob (packing, reaching, and grasping) [23], we organize the evaluation into three tasks of increasing interaction complexity. Given an initial RGB observation and a language instruction, SWIM must perform the following tasks:

- **Packing.** The instruction specifies a curling direction. The robot must deform its body into a compact configuration in the requested direction.
- **Reaching.** The instruction specifies a target object in the visual scene. The robot must deform its body to bring its distal portion close to the target.
- **Grasping.** The instruction specifies a target object in the visual scene. The robot must approach and wrap its body around the target.

TABLE I
IMPLEMENTATION AND TRAINING DETAILS OF SWIM-VLA.

Setting	Value
VLA backbone	OpenVLA-OFT [25].
Policy observation	Top-down RGB image, language instruction, and current tendon-length vector.
Image resolution	256×256 ; OpenVLA-OFT image preprocessing.
Action space	Absolute motor-torque commands $\mathbf{a}_t \in \mathbb{R}^2$ for the two tendon drives, using the same torque-space interface in simulation and physical execution.
Action normalization	Percentile-bounded normalization using the dataset 1st and 99th percentiles.
Command rate	10 Hz.
Prediction horizon	$H = 15$ commands, corresponding to 1.5 s.
Execution horizon	$K = 8$ commands (0.8 s) per policy query.
Diffusion training	50 diffusion steps with a noise-prediction objective.
Diffusion inference	10 DDIM denoising steps.
VSP targets	$N_b = 4$ ordered body anchors selected along the backbone at approximately one-quarter, one-half, three-quarters, and the distal tip.
VSP head	Two-layer MLP with a 512-dimensional hidden layer, GELU activation, and linear output.
VSP loss	Mean Euclidean localization error over the N_b ordered body anchors.
VSP loss weight	$\lambda_{\text{VSP}} = 0.15$.
Fine-tuning	LoRA with rank 32 and bfloat16 mixed precision.
Optimizer	AdamW.
Learning rate	3×10^{-4} with cosine decay.
Batch size	4 samples per GPU.
Training budget	20,000 optimization steps.
Compute	Four NVIDIA A100 GPUs.
Policy inference time	Approximately 700 ms per query (mean).

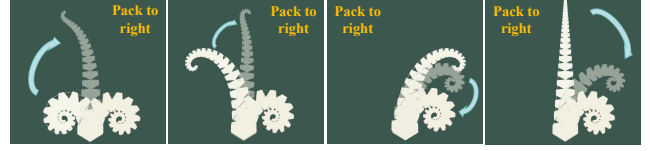
Task success is assessed from the final robot configuration in each simulated rollout and physical test. A packing trial is considered successful when the robot reaches a fully curled configuration in the instructed direction. A reaching trial is considered successful when the robot’s distal portion is close to the language-specified object in the final configuration. A grasping trial is considered successful when the robot wraps its body around the language-specified target.

3) *Demonstration dataset*: We generate actuation-space demonstrations in MuJoCo following the expert-demonstration procedure of prior whole-body grasping work [10] and the characteristic motion patterns of SpiRob [23]. Each training condition defines a task setup, for which we collect four different successful trajectories with distinct actuation-command sequences. Each trajectory contains synchronized RGB observations, language instructions, tendon lengths, and actuation commands, together with ordered body-anchor coordinates for VSP supervision. For packing, a training condition is defined by the initial robot configuration and the instructed curling direction. For reaching, the instruction specifies a target object in the scene for the robot to approach. For grasping, training conditions vary in target-object position, color (red, green, and blue), and shape (cylinder, pentagram, rectangular prism, and hexagonal prism). Reaching and grasping scenes also contain visual distractors; only

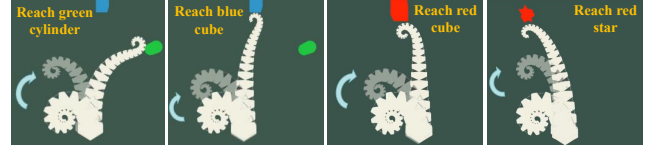
TABLE II
SIMULATION DEMONSTRATION AND EVALUATION DATASET.

Task	Train cond.	Traj. per cond.	Train traj.
Packing	18	4	72
Reaching	32	4	128
Grasping	148	4	592

A. Different images but same language Instruction (Packing)



B. Same images but different language Instruction (Reaching)



C. Same images but different language Instruction (Grasping)

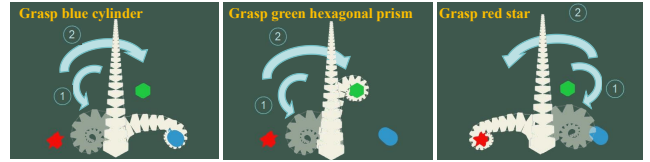


Fig. 3. Qualitative evaluation of SWIM-VLA.

the language-specified target participates in contact and dynamics. During grasping, target objects remain rigidly anchored at their sampled planar poses. Table II summarizes the training conditions and corresponding demonstration trajectories.

B. Simulation Evaluation

We evaluate all methods on held-out test conditions not used for training, including 50 packing, 50 reaching, and 100 grasping conditions, with one rollout per condition.

a) *Compared methods and ablations*: We compare the following complete policies and controlled SWIM-VLA variants. All methods share the same backbone, inputs, data, and training and evaluation settings.

- **OpenVLA-OFT**. We adopt OpenVLA-OFT [13], one of the VLA policies evaluated in ManiSoft [15], and adapt it to the same actuation-command space and multimodal inputs as SWIM-VLA. The baseline retains its deterministic L_1 action-prediction objective.
- **SWIM-VLA (Ours)**. The complete SWIM-VLA policy combines VSP supervision with a conditional diffusion action head.
- **SWIM-VLA without VSP**. The auxiliary whole-body geometric objective is disabled by setting $\lambda_{\text{VSP}} = 0$.
- **SWIM-VLA without a diffusion action head**. The conditional diffusion action head is replaced with a deterministic H -step regressor trained using an L_1 objective.

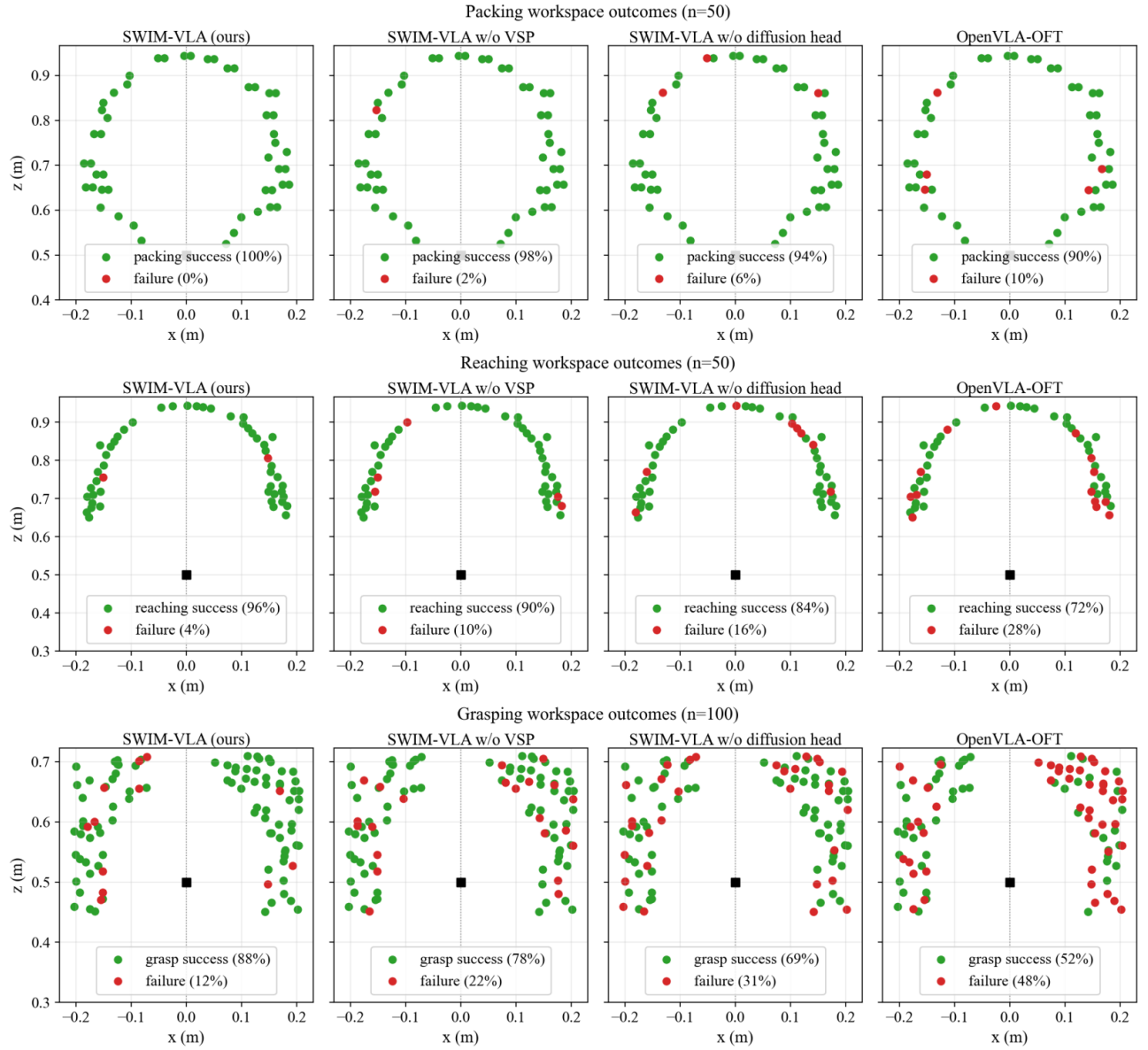


Fig. 4. Spatial distributions of simulation outcomes. Marker locations represent initial positions for packing and target-object positions for reaching and grasping. Green and red markers indicate successful and failed trials, respectively.

TABLE III
SIMULATION SUCCESS RATES (%) ON HELD-OUT TEST CONDITIONS.

Method	Packing	Reaching	Grasping
OpenVLA-OFT	90	72	52
SWIM-VLA(Ours)	100	96	88
w/o VSP	98	90	78
w/o diffusion action head	94	84	69

b) Simulation results: Table III reports success rates of 100%, 96%, and 88% for SWIM-VLA on packing, reaching, and grasping, respectively, compared with 90%, 72%, and 52% for OpenVLA-OFT. The larger improvements between the two methods on reaching and grasping suggest particular

benefits for target-directed deformation and external contact coordination. Fig. 3(A) shows different motions from different initial configurations under the same packing instruction, illustrating the realization of a shared goal from different states. Panel (B) shows the robot approaching the language-specified object, while panel (C) shows that changing the instruction within the same initial scene redirects target selection and wrapping. Together, these examples show how a jointly trained policy adapts whole-body behavior to the current configuration and instructed goal. Fig. 4 further reveals the spatial distribution of task outcomes. Packing remains robust across the sampled initial positions, whereas reaching and grasping show greater sensitivity to target location. Failures become more frequent toward peripheral target regions when VSP or diffusion is removed and are

most widespread for OpenVLA-OFT. In contrast, SWIM-VLA maintains more consistent success across the evaluated workspace. Removing VSP lowers reaching and grasping success rates to 90% and 78%, while packing remains at 98% (Table III), supporting the value of geometric supervision particularly for distal placement and contact formation. Replacing diffusion with deterministic $L1$ regression reduces grasping success from 88% to 69%, with smaller degradations in packing and reaching. Since each training condition contains multiple successful trajectories with distinct actuation-command sequences (Table II), the diffusion head is better suited to modeling the corresponding conditional action distribution than a single deterministic prediction. The larger degradation in grasping further suggests that contact-rich whole-body interaction is more sensitive to the quality of the predicted actuation sequence.

C. Physical Evaluation

We validate the complete SWIM framework on the physical robot using the trained SWIM-VLA policy. For each task, we conduct 20 physical trials for each deployment strategy. The studied robot (Fig. 5) operates in a planar workspace observed by a fixed overhead RGB camera. Additional non-target distractors remain visible in the camera view but do not participate in physical interactions.

a) Compared deployment strategies: We compare two deployment strategies using the same trained SWIM-VLA checkpoint.

- **Direct SWIM-VLA.** The policy is queried online using physical RGB observations and tendon-length vectors. At each query, it predicts a 15-step command chunk, of which the first 8 commands are executed before the next policy query.
- **SWIM.** The initial physical observation is first used to initialize the virtual scene. Task objects are segmented in HSV space, and their image centroids are mapped to the MuJoCo workspace using a one-time planar calibration [24]. Predefined object models with measured dimensions and matching colors are then placed at the estimated planar locations. The same SWIM-VLA policy performs iterative rollouts with the evolving simulated state, and the executed commands are collected into a complete actuation-command sequence. The resulting sequence is then executed open-loop on

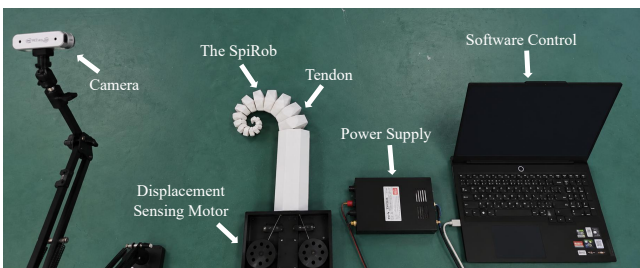


Fig. 5. Physical deployment

A. “Grasp green hexagonal prism” (Direct SWIM-VLA)



B. “Grasp green hexagonal prism” (SWIM)



Fig. 6. Physical grasping under the instruction “Grasp green hexagonal prism” using the same trained SWIM-VLA checkpoint. (A) Direct online deployment. (B) SWIM with commands constructed by virtual rollout.

the physical robot without online policy inference or replanning.

TABLE IV

PHYSICAL TASK SUCCESS COUNTS FOR DIRECT SWIM-VLA DEPLOYMENT AND THE COMPLETE SWIM FRAMEWORK.

Method	Packing	Reaching	Grasping
Direct SWIM-VLA	15/20	8/20	5/20
SWIM	20/20	16/20	15/20

b) Physical deployment results: SWIM achieves success rates of 100%, 80%, and 75% on packing, reaching, and grasping, respectively (Table IV), substantially outperforming the direct deployment of the same SWIM-VLA checkpoint, which achieves 75%, 40%, and 25%. For reference, SWIM-VLA achieves success rates of 100%, 96%, and 88% in simulation (Table III). Thus, SWIM recovers a substantial portion of the performance lost under direct physical deployment, although reaching and grasping remain below their simulation results. Fig. 6 illustrates incomplete enclosure under direct deployment but successful wrapping with SWIM. The degradation under direct deployment is consistent with sim-to-real mismatch and online inference latency (Table I). Motivated by the robot’s embodied mechanical intelligence, SWIM shifts policy interaction to the virtual scene and executes the resulting command sequence open-loop on the physical robot. Intrinsic body compliance supports local contact adaptation, reducing the need for repeated real-image feedback and online VLA inference during execution. The remaining gap relative to simulation may arise from object-position errors during scene initialization and differences in the robot’s material and structural properties, which can cause the same commands to produce different physical deformations.

V. CONCLUSION

In this paper, we present SWIM, a framework for vision-language-grounded whole-body manipulation with

soft robots. SWIM-VLA combines actuation-space diffusion prediction with Visual Soft Proprioception to learn language-conditioned whole-body commands from limited simulated demonstrations. At deployment, SWIM constructs a complete command sequence through virtual rollouts and executes it open-loop on the physical robot, where body compliance supports local contact adaptation. Simulation results show the benefits of both VSP and diffusion-based action prediction, while physical experiments demonstrate that SWIM substantially outperforms the direct online deployment of the same SWIM-VLA checkpoint and recovers much of the performance lost during direct sim-to-real transfer. The current study is limited to planar manipulation, predefined reset configurations, and anchored grasping targets. Future work will consider more general 3D interactions, moving objects, and improved scene and physical-model alignment.

REFERENCES

- [1] D. Rus and M. T. Tolley, "Design, fabrication and control of soft robots," *Nature*, vol. 521, no. 7553, pp. 467–475, 2015.
- [2] I. D. Walker, "Continuous backbone "continuum" robot manipulators," *ISRN Robotics*, vol. 2013, pp. 1–19, 2013.
- [3] W. McMahan, V. Chitrakaran, M. Csencsits, D. Dawson, I. D. Walker, B. A. Jones, M. Pritts, D. Dienno, M. Grissom, and C. D. Rahn, "Field trials and testing of the OctArm continuum manipulator," in *Proceedings of the 2006 IEEE International Conference on Robotics and Automation*, 2006, pp. 2336–2341.
- [4] J. Li and J. Xiao, "Progressive planning of continuum grasping in cluttered space," *IEEE Transactions on Robotics*, vol. 32, no. 3, pp. 707–716, 2016.
- [5] P. Pustina, P. Borja, C. Della Santina, and A. De Luca, "P-satl-D shape regulation of soft robots," *IEEE Robotics and Automation Letters*, vol. 8, no. 1, pp. 1–8, 2023.
- [6] E. Almazor, F. Ye, J. Shi, T. G. Thuruthel, H. A. Würdemann, and F. Iida, "Static shape control of soft continuum robots using deep visual inverse kinematic models," *IEEE Transactions on Robotics*, vol. 39, no. 4, pp. 2973–2988, 2023.
- [7] A. D. Marchese, R. K. Katzschmann, and D. Rus, "Whole arm planning for a soft and highly compliant 2d robotic manipulator," in *2014 IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2014, pp. 554–560.
- [8] M. A. Graule, C. B. Teeple, and R. J. Wood, "Contact-implicit trajectory and grasp planning for soft continuum manipulators," in *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2022, pp. 9401–9408.
- [9] E. Del Dottore, R. Adhami, E. Shahabi, E. Solfiti, M. Martini, S. Mariani, A. Parmiggiani, A. Mondini, E. Sinibaldi, and B. Mazzolai, "Peripheral control enabled by distributed sensing in an octopus-inspired soft robotic arm for autonomous underwater grasping," *Nature Machine Intelligence*, vol. 8, pp. 708–721, 2026.
- [10] L. Yang, Y. Bai, Y. Wang, I. Alsarraj, G. Kutyniok, Z. Wang, and K. Wu, "Lightweight learning from actuation-space demonstrations via flow matching for whole-body soft robotic grasping," *IEEE Robotics and Automation Letters*, vol. 11, no. 6, pp. 6720–6727, 2026.
- [11] A. Brohan, N. Brown, J. Carbajal, Y. Chebotar, X. Chen, K. Choremanski, T. Ding, D. Driess, A. Dubey, C. Finn, P. Florence, C. Fu, M. Gonzalez Arenas, K. Gopalakrishnan, K. Han, K. Hausman, A. Herzog, J. Hsu, B. Ichter, A. Irpan, N. Joshi, R. Julian, D. Kalashnikov, Y. Kuang, I. Leal, L. Lee, T.-W. E. Lee, S. Levine, Y. Lu, H. Michalewski, I. Mordatch, K. Pertsch, K. Rao, K. Reymann, M. Ryoo, G. Salazar, P. Sanketi, P. Sermanet, J. Singh, A. Singh, R. Soricut, H. Tran, V. Vanhoucke, Q. Vuong, A. Wahid, S. Welker, P. Wohlhart, J. Wu, F. Xia, T. Xiao, P. Xu, S. Xu, T. Yu, and B. Zitkovich, "RT-2: Vision-language-action models transfer web knowledge to robotic control," *arXiv preprint arXiv:2307.15818*, 2023.
- [12] M. J. Kim, K. Pertsch, S. Karamcheti, T. Xiao, A. Balakrishna, S. Nair, R. Rafailov, E. Foster, G. Lam, P. Sanketi *et al.*, "Open-vla: An open-source vision-language-action model," *arXiv preprint arXiv:2406.09246*, 2024.
- [13] M. J. Kim, C. Finn, and P. Liang, "Fine-tuning vision-language-action models: Optimizing speed and success," *arXiv preprint arXiv:2502.19645*, 2025.
- [14] H. Su, C. Meo, F. Stella, A. Peirone, K. Junge, and J. Hughes, "Bridging embodiment gaps: Deploying vision-language-action models on soft robots," *arXiv preprint arXiv:2510.17369*, 2025.
- [15] Z. Wei, L. Wang, C. Gao, L. Wen, and S. Liu, "ManiSoft: Towards vision-language manipulation for soft continuum robotics," *arXiv preprint arXiv:2605.18617*, 2026.
- [16] C. K. Ng, L. Bai, G. Wang, Y. Wang, H. Gao, K. Yuan, C. Jin, T. Zeng, and H. Ren, "EndoVLA: Dual-phase vision-language-action model for autonomous tracking in endoscopy," *arXiv preprint arXiv:2505.15206*, 2025.
- [17] J. Lin, Z. Xiong, Z. Pan, Y. Hu, T. Tamima, Z. Hao, E. Cheung, J. Lai, H. Gao, H. Ren *et al.*, "Bilivla: Scene-aware vision-language-action model with reinforcement learning for autonomous biliary endoscopic navigation," *arXiv preprint arXiv:2606.23531*, 2026.
- [18] R. Tang, C. K. Ng, K. Wu, L. Bai, G. Wang, Y. Huang, Y. Wang, and H. Ren, "TMR-VLA: Vision-language-action model for magnetic motion control of tri-leg silicone-based soft robot," *arXiv preprint arXiv:2603.00420*, 2026.
- [19] T. Liu, T. Chen, S. Ma, Y. Wang, A. P. P. Aung, I. Alsarraj, J. Senthilnath, B. An, and K. Wu, "Whole-body semantic-to-actuation grounding of elephant-inspired soft-trunk motion via lightweight flow matching," *arXiv preprint arXiv:2607.11018*, 2026.
- [20] C. Chi, Z. Xu, S. Feng, E. Cousineau, Y. Du, B. Burchfiel, R. Tedrake, and S. Song, "Diffusion policy: Visuomotor policy learning via action diffusion," *The International Journal of Robotics Research*, vol. 44, no. 10–11, pp. 1684–1704, 2025.
- [21] P. Rao, Q. Peyron, S. Lilge, and J. Burgner-Kahrs, "How to model tendon-driven continuum robots and benchmark modelling performance," *Frontiers in Robotics and AI*, vol. 7, p. 630245, 2021.
- [22] K. Black, M. Galliker, and S. Levine, "Real-time execution of action chunking flow policies," in *Advances in Neural Information Processing Systems*, vol. 38, 2025, pp. 37 596–37 620.
- [23] Z. Wang, N. M. Freris, and X. Wei, "SpiRobs: Logarithmic spiral-shaped robots for versatile grasping across scales," *Device*, vol. 3, no. 4, p. 100646, 2025.
- [24] Z. Zhang, "A flexible new technique for camera calibration," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 22, no. 11, pp. 1330–1334, 2000.
- [25] M. J. Kim, C. Finn, and P. Liang, "Fine-tuning vision-language-action models: Optimizing speed and success," *arXiv preprint arXiv:2502.19645*, 2025.